

Supplementary Material for “Minimax Random Reshuffling in Every Convex Regime”

1 Notation and a Permutation Estimate

We first prove the all-iterate asymmetric bridge. Let $n \geq 32$ be even. Let $s = (s_1, \dots, s_n)$ be a uniform random permutation of $n/2$ signs equal to $+1$ and $n/2$ signs equal to -1 . Define $S_i = \sum_{j=1}^i s_j$. Let

$$h_\ell(x) = \frac{\ell}{2}(x_-)^2 + \frac{\mu_0}{2}(x_+)^2, \quad g(x) = h'_\ell(x), \quad \mu_0 = \frac{\ell}{2415}. \quad (1)$$

We write $h = h_\ell$ when the curvature is clear. One epoch of the active coordinate follows

$$v_{i+1} = v_i - \eta(g(v_i) + \sigma s_{i+1}), \quad i = 0, \dots, n-1. \quad (2)$$

Write $z = \eta\ell$ and assume $zn \leq 1/161$.

We use the following estimates from the proof of Theorem 3.1 of Cha et al. (2023). We state the specialization needed here.

Lemma 1 (One-epoch estimates). *For a fixed start $v_0 = y \geq 0$, the process (2) satisfies*

$$\mathbb{E}g(v_i) \leq \frac{2}{3}\ell y - \frac{\eta\ell\sigma}{480}\sqrt{i}, \quad 0 \leq i \leq n/2, \quad (3)$$

$$\mathbb{E}g(v_i) \leq \left(1 + \frac{161}{160}iz\right)\mu_0 y + \frac{161}{160}\eta\mu_0\sigma\sqrt{i}, \quad 0 \leq i \leq n-1. \quad (4)$$

If the complete multi-epoch process starts at zero, every epoch boundary Y_k satisfies

$$\Pr(Y_k \geq 0) \geq \frac{1}{2}. \quad (5)$$

The first inequality is Lemma B.2 of Cha et al. (2023). It follows by conditioning on the sign of S_i , using $\mathbb{E}|S_i| \asymp \sqrt{i}$, and controlling the difference $v_i - y$ by a discrete Gronwall inequality. The second is their Lemma B.3, and the boundary sign probability is their Lemma B.4. Their assumptions hold because $\ell/\mu_0 = 2415$ and $zn \leq 1/161$.

We also need a direct pathwise comparison. For any fixed start $y \in \mathbb{R}$, couple (2) to

$$q_{i+1} = (1-z)q_i - \eta\sigma s_{i+1}, \quad q_0 = y. \quad (6)$$

The map $x \mapsto x - \eta g(x)$ is increasing because $z < 1$, and it is at least $(1-z)x$ for every x . Induction under the common sign sequence gives $v_i \geq q_i$ for every i .

For clarity, the movement estimate underlying (3)–(4) is

$$\mathbb{E}|v_i - y| \leq \frac{161}{160} \left(\eta\ell i y + \eta\sigma\sqrt{i} \right). \quad (7)$$

Indeed, telescoping (2), using $|g(v)| \leq \ell|v|$, and the without-replacement identity $\mathbb{E}S_i^2 = i(n-i)/(n-1) \leq i$ gives

$$\mathbb{E}|v_i - y| \leq \eta\ell i y + \eta\ell \sum_{j=0}^{i-1} \mathbb{E}|v_j - y| + \eta\sigma\sqrt{i}.$$

Since $zn \leq 1/161$, discrete Gronwall gives (7). Substitution directly proves (4). The sharper signed estimate (3) also uses $\mathbb{E}|S_i| \geq \sqrt{i}/10$ and $\Pr(S_i > 0) = \Pr(S_i < 0) \geq 1/6$ for $i \leq n/2$. These are the only external permutation facts used below.

2 Proof of the All-Iterate Bridge

We now prove the all-iterate bridge lemma from the main paper, with all index and constant details.

2.1 Fixed nonnegative start

For a fixed $y \geq 0$, summing (2) and using $\mathbb{E}S_i = 0$ gives

$$\mathbb{E}[v_i \mid v_0 = y] = y - \eta \sum_{j=0}^{i-1} \mathbb{E}g(v_j). \quad (8)$$

For $1 \leq i \leq n/2 + 1$, substituting (3) yields

$$\mathbb{E}[v_i \mid y] \geq \left(1 - \frac{2}{3}zi\right)y + \frac{\eta^2 \ell \sigma}{480} \sum_{j=0}^{i-1} \sqrt{j}. \quad (9)$$

For $n/2 + 1 < i \leq n$, use (3) through $j = n/2$ and (4) afterward. This gives

$$\mathbb{E}[v_i \mid y] \geq (1 - \eta A_i)y + \eta^2 \ell \sigma B_i, \quad (10)$$

where

$$A_i = \frac{2}{3}\ell \left(\frac{n}{2} + 1\right) + \mu_0 \sum_{j=n/2+1}^{i-1} \left(1 + \frac{161}{160}jz\right), \quad (11)$$

$$B_i = \frac{1}{480} \sum_{j=0}^{n/2} \sqrt{j} - \frac{161}{160 \cdot 2415} \sum_{j=n/2+1}^{i-1} \sqrt{j}. \quad (12)$$

Because $jz \leq nz \leq 1/161$ and $n \geq 32$,

$$\begin{aligned} \frac{A_i}{\ell n} &\leq \frac{1}{3} + \frac{2}{3n} + \frac{1}{2 \cdot 2415} \frac{161}{160} \\ &\leq 0.354362 < 0.36. \end{aligned} \quad (13)$$

The adverse part of B_i increases with i . Hence

$$\begin{aligned} \frac{B_i}{n^{3/2}} &\geq \frac{1}{480n^{3/2}} \int_0^{n/2} \sqrt{t} dt - \frac{161}{160 \cdot 2415} \frac{1}{2} \\ &= \frac{1}{1440\sqrt{2}} - \frac{161}{320 \cdot 2415} > \frac{1}{18000}. \end{aligned} \quad (14)$$

We used at most $n/2$ adverse summands, each bounded by $\sqrt{n}$. Combining (10)–(14),

$$\mathbb{E}[v_i \mid y] \geq (1 - 0.36\alpha)y + \frac{\eta^2 \ell \sigma n^{3/2}}{18000}, \quad \alpha = zn. \quad (15)$$

2.2 A common coefficient for negative starts

For $y < 0$, Lemma 1 and marginal symmetry of each sign give

$$\mathbb{E}[v_i \mid y] \geq \mathbb{E}[q_i \mid y] = (1 - z)^i y. \quad (16)$$

For $u \in [0, 1/161]$,

$$e^{-u} \leq 1 - u + \frac{u^2}{2} \leq 1 - \frac{321}{322}u. \quad (17)$$

When $i \leq n/2 + 1$, this implies

$$(1 - z)^i \leq e^{-zi} \leq 1 - \frac{2}{3}zi. \quad (18)$$

When $i > n/2 + 1$, we have $zi > \alpha/2$, so

$$(1 - z)^i \leq e^{-zi} \leq 1 - \frac{321}{644}\alpha \leq 1 - 0.36\alpha. \quad (19)$$

Because $y < 0$, multiplying (18) or (19) by y reverses the inequality. Thus the coefficients in (9) and (15) also lower bound the negative-start conditional means.

2.3 Random epoch starts and complete averaging

Let $Y_k = v_{k,0}$ and $M_k = \mathbb{E}Y_k$. The process starts with $Y_1 = 0$. By (5), $\Pr(Y_k \geq 0) \geq 1/2$. Averaging the common-coefficient bounds over positive and negative starts gives, for every pre-update index $0 \leq i \leq n-1$,

$$\mathbb{E}v_{k,i} \geq (1 - 0.36\alpha)M_k. \quad (20)$$

For the first-half indices we used $(2/3)zi \leq (2/3)(1/2 + 1/n)\alpha < 0.36\alpha$. For $n/2 + 2 \leq i \leq n-1$, the positive-start event also contributes the drift in (15), so

$$\mathbb{E}v_{k,i} \geq (1 - 0.36\alpha)M_k + \frac{\alpha q}{36000}, \quad q = \eta\sigma\sqrt{n}. \quad (21)$$

The same inequality holds at the endpoint $i = n$. Therefore

$$M_{k+1} \geq (1 - \beta)M_k + \gamma, \quad \beta = 0.36\alpha, \quad \gamma = \frac{\alpha q}{36000}. \quad (22)$$

In particular, $M_k \geq 0$ for every k , which justifies dropping the nonnegative common-coefficient terms when needed.

Unrolling (22) gives

$$M_k \geq \frac{\gamma}{\beta} [1 - (1 - \beta)^{k-1}]. \quad (23)$$

For every integer $j \geq 0$,

$$1 - (1 - \beta)^j \geq 1 - e^{-\beta j} \geq (1 - e^{-1}) \min\{\beta j, 1\}. \quad (24)$$

For $K \geq 2$, monotonicity of the right side over the final half of its indices implies

$$\frac{1}{K} \sum_{k=1}^K M_k \geq c_0 q \min\{(K-1)\alpha, 1\} \quad (25)$$

for a numerical $c_0 > 0$.

There are $n/2 - 2 \geq n/3$ pre-update indices covered by (21). Averaging (20) and (21) over all nK output iterates gives

$$\begin{aligned} \mathbb{E}\bar{v} &\geq (1 - 0.36\alpha) \frac{1}{K} \sum_{k=1}^K M_k + \frac{\alpha q}{108000} \\ &\geq c_1 q \min\{K\alpha, 1\}. \end{aligned} \quad (26)$$

The direct second term covers $K = 1$. Since

$$h(v) \geq \frac{\ell}{2 \cdot 2415} v^2, \quad (27)$$

Jensen's inequality yields

$$\mathbb{E}h(\bar{v}) \geq c_2 \ell \eta^2 n \sigma^2 \min\{K\eta\ell n, 1\}^2. \quad (28)$$

This proves the all-iterate bridge lemma.

3 Proof of the Fixed-Step Theorem

We give the complete direct-sum argument for even n . Let $T = nK$ and set

$$\lambda = \min \left\{ L, \frac{1}{2\eta T} \right\}, \quad \ell = \min \left\{ L, \frac{1}{161\eta n} \right\}. \quad (29)$$

Take balanced $b_i \in \{-\sigma, +\sigma\}$ and define

$$f_i(u, v, w) = \frac{\lambda}{2} u^2 + h(v) + b_i v + \frac{L}{2} w^2. \quad (30)$$

Initialize at $(D, 0, 0)$. The average objective has unique minimizer zero. Every component is convex. Its Hessian slopes belong to $\{\lambda, \ell, \ell/2415, L\}$, so its minimal gradient-Lipschitz constant is exactly L . The initial distance is D , and

$$\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(0)\|^2 = \frac{1}{n} \sum_{i=1}^n b_i^2 = \sigma^2. \quad (31)$$

The dormant coordinate stays zero. In the first coordinate, the global pre-update sequence is $u_t = (1 - \eta\lambda)^t D$. The base is nonnegative because $\eta L \leq 1/6$. Bernoulli's inequality and $\eta\lambda T \leq 1/2$ give

$$u_t \geq (1 - \eta\lambda t)D \geq D/2, \quad 0 \leq t < T. \quad (32)$$

Consequently,

$$\frac{\lambda}{2} \bar{u}^2 \geq \frac{\lambda D^2}{8} \geq \frac{1}{16} \min \left\{ LD^2, \frac{D^2}{\eta T} \right\}. \quad (33)$$

The second coordinate starts at zero and satisfies the bridge assumptions. From (28),

$$\mathbb{E}h(\bar{v}) \geq c_2 \ell \eta^2 n \sigma^2 \min\{K\eta \ell n, 1\}^2. \quad (34)$$

Also,

$$\ell \eta^2 n = \min\{\eta^2 n L, \eta/161\} \geq \frac{1}{161} \min\{\eta^2 n L, \eta\}. \quad (35)$$

Suppose first that $\eta L T \geq 1/161$. If $\eta n L < 1/161$, then $K\eta \ell n = K\eta n L = \eta L T \geq 1/161$. If $\eta n L \geq 1/161$, then $K\eta \ell n = K/161 \geq 1/161$. Thus the last factor of (34) is at least $1/161^2$. Combining (33)–(35) gives

$$\mathbb{E}[f(\bar{x}) - f(0)] \geq c_3 \left[\min \left\{ LD^2, \frac{D^2}{\eta T} \right\} + \min\{\eta^2 n L, \eta\} \sigma^2 \right]. \quad (36)$$

The right side lower bounds a constant multiple of

$$\min \left\{ LD^2, \frac{D^2}{\eta T} + \min\{\eta^2 n L, \eta\} \sigma^2 \right\}. \quad (37)$$

If $\eta L T < 1/161$, then $\lambda = L$, and (33) is at least $LD^2/8$. This directly lower bounds (37). The fixed-step theorem follows.

4 Odd Component Counts

Let $n \geq 33$ be odd and $m = n - 1$. Use the preceding even construction on m active components. Add one dummy component that is zero in the u and v coordinates. Include $Lw^2/2$ in all n components, including the dummy. Set the active sign magnitudes to

$$\sigma_a = \sigma \sqrt{\frac{n}{m}}. \quad (38)$$

Then the full instance has $\sigma_\star = \sigma$ exactly. Its average objective in the active coordinates is m/n times the even objective.

Deleting the dummy from a uniform permutation of $[n]$ leaves a uniform permutation of the m active components. The dummy update does not change the active state. The n pre-update states in an epoch are the m active pre-update states plus one repeated active state. Conditional on the rank of the dummy, this repeated state is v_i for some $0 \leq i \leq m$. Every such state has nonnegative expectation by (20) and the endpoint recursion. Therefore

$$\mathbb{E}\bar{v}_n \geq \frac{m}{n} \mathbb{E}\bar{v}_m. \quad (39)$$

The deterministic coordinate is positive pathwise, so the same comparison holds there. Apply the even construction with

$$\ell = \min\{L, 1/(161\eta m)\}, \quad \lambda = \min\{L, 1/(2\eta m K)\}. \quad (40)$$

Since $m/n \geq 32/33$ and m is within a fixed factor of n , all target terms change by at most universal constants. This proves the theorem for odd n .

5 Heterogeneous Smoothness

We prove Theorem 4 from the main paper. Let

$$q_{\text{eff}} = \frac{n\bar{L}}{\hat{L}} \geq 34.$$

Choose an even integer m as follows. If $q_{\text{eff}} = n$ and n is even, take $m = n$. Otherwise take the largest even integer strictly below q_{eff} . Then

$$q_{\text{eff}} - 2 \leq m < q_{\text{eff}}, \quad \frac{16}{17}\bar{L} \leq \frac{m\hat{L}}{n} \leq \bar{L}. \quad (41)$$

In particular, $m \geq 32$.

There are m active components and $n - m$ inactive components. Define

$$\delta = \frac{n\bar{L} - m\hat{L}}{n - m} \quad (42)$$

when $m < n$. The strict choice of m gives $0 < \delta \leq \hat{L}$. On a common dormant coordinate w , active components contain $\hat{L}w^2/2$ and inactive components contain $\delta w^2/2$. Thus the maximum component smoothness is exactly $\hat{L}$ and the average is exactly $\bar{L}$. When $m = n$, all components contain $\hat{L}w^2/2$ and $\bar{L} = \hat{L}$.

Set

$$\sigma_a = \sigma \sqrt{\frac{n}{m}}, \quad \lambda = \min \left\{ \hat{L}, \frac{1}{2\eta m K} \right\}, \quad \ell = \min \left\{ \hat{L}, \frac{1}{161\eta m} \right\}. \quad (43)$$

For the active components, use

$$f_i(u, v, w) = \frac{\lambda}{2}u^2 + h_\ell(v) + b_i v + \frac{\hat{L}}{2}w^2, \quad b_i \in \{-\sigma_a, +\sigma_a\}, \quad (44)$$

with balanced signs. For inactive components use $f_i(u, v, w) = \delta w^2/2$. Initialize at $(D, 0, 0)$. Every component is convex. The active components have minimal smoothness $\hat{L}$, the inactive components have minimal smoothness δ , and

$$\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(0)\|^2 = \frac{m}{n} \sigma_a^2 = \sigma^2. \quad (45)$$

The average objective in the hard coordinates is

$$f(u, v, 0) = \frac{m}{n} \left(\frac{\lambda}{2}u^2 + h_\ell(v) \right). \quad (46)$$

5.1 Uniform interleaving

Delete the inactive entries from a uniform permutation of $[n]$. The remaining active order is uniform, and conditional on this order the inactive entries are uniformly interleaved. Let d_r be the number of inactive entries after r active updates and before active update $r + 1$, where $0 \leq r < m$. Let d_m count inactive entries after all active updates. Symmetry gives

$$\mathbb{E}d_r = \frac{n - m}{m + 1}, \quad 0 \leq r \leq m. \quad (47)$$

Let $z_0, \dots, z_m$ be one coordinate's active states. The full output in that epoch contains $d_r + 1$ copies of z_r for $r < m$, and d_m copies of z_m . The interleaving is independent of the active sign order. Therefore

$$\mathbb{E}[\bar{z}_{\text{full}} \mid z_0, \dots, z_m] = \frac{n + 1}{n(m + 1)} \sum_{r=0}^{m-1} z_r + \frac{n - m}{n(m + 1)} z_m. \quad (48)$$

After averaging over active orders, every bridge state and endpoint has nonnegative mean. It follows that

$$\mathbb{E}\bar{z}_{\text{full}} \geq \frac{m(n + 1)}{n(m + 1)} \mathbb{E}\bar{z}_{\text{active}} \geq \frac{1}{2} \mathbb{E}\bar{z}_{\text{active}}. \quad (49)$$

For the deterministic coordinate, all states are at least $D/2$ pathwise.

5.2 Fixed-step lower bound

The deterministic active process makes mK updates, and $\eta\lambda mK \leq 1/2$. By (46),

$$\begin{aligned} \frac{m}{n} \frac{\lambda}{2} \bar{u}^2 &\geq c \min \left\{ \frac{m\hat{L}}{n} D^2, \frac{D^2}{\eta m K} \right\} \\ &\geq c' \min \left\{ \bar{L} D^2, \frac{D^2}{\eta m K} \right\}. \end{aligned} \quad (50)$$

Apply the bridge lemma to the m active components with variance σ_a^2 . Equations (46) and (49) give

$$\begin{aligned} \mathbb{E}f(0, \bar{v}, 0) &\geq c \frac{m}{n} \ell \eta^2 m \sigma_a^2 \min\{K\eta\ell m, 1\}^2 \\ &= c \ell \eta^2 m \sigma^2 \min\{K\eta\ell m, 1\}^2. \end{aligned} \quad (51)$$

By (41),

$$\ell \eta^2 m = \min\{\eta^2 m \hat{L}, \eta/161\} \asymp \min\{\eta^2 n \bar{L}, \eta\}. \quad (52)$$

If $\eta m \hat{L} K$ is at least a numerical constant, the attenuation in (51) is a numerical constant. Otherwise $\lambda = \hat{L}$, and (50) is $\Omega(\bar{L} D^2)$. The case argument from Section 3 proves

$$\mathbb{E}[f(\bar{x}) - f(0)] \geq c \min \left\{ \bar{L} D^2, \frac{D^2}{\eta m K} + \min\{\eta^2 n \bar{L}, \eta\} \sigma^2 \right\}. \quad (53)$$

The scalar optimization below applies with the noise curvature L replaced by $\bar{L}$ and the maximum step restricted by $\hat{L}$. The condition $n\bar{L}/\hat{L} \geq 34$ places the transition $1/(n\bar{L})$ below the endpoint $1/(6\hat{L})$ by a fixed factor. It yields

$$\inf_{\eta} \left[\frac{D^2}{\eta m K} + \min\{\eta^2 n \bar{L}, \eta\} \sigma^2 \right] \asymp \frac{\hat{L} D^2}{n K} + \min \left\{ \frac{\sigma D}{\sqrt{n K}}, \left(\frac{\bar{L} \sigma^2 D^4}{n K^2} \right)^{1/3} \right\}. \quad (54)$$

Together with (53), this proves the heterogeneous minimax lower bound. The matching upper bound is due to Liu (2026), plus the $\bar{L} D^2/2$ small-step cap.

6 Uniform Small-Step Cap

We justify the smoothness cap uniformly over the minimax class. Let $a_i = \|\nabla f_i(x_*)\|$ and let L_i be valid component smoothness constants. For any realized sequence of component indices j_t , write $r_t = \|x_t - x_*\|$. Smoothness gives

$$r_{t+1} \leq (1 + \eta L_{j_t}) r_t + \eta a_{j_t}. \quad (55)$$

Across K complete permutations,

$$\sum_{t=0}^{nK-1} L_{j_t} = nK \bar{L}, \quad \sum_{t=0}^{nK-1} a_{j_t} = K \sum_{i=1}^n a_i \leq nK \sigma. \quad (56)$$

Unrolling the recursion and using $1 + u \leq e^u$ yields the pathwise bound

$$\max_{0 \leq t < nK} r_t \leq e^{\eta n K \bar{L}} (D + \eta n K \sigma). \quad (57)$$

The average objective is $\bar{L}$ -smooth and has zero gradient at x_* . Convexity of the norm and (57) therefore give

$$f(\bar{x}) - f(x_*) \leq \frac{\bar{L}}{2} e^{2\eta n K \bar{L}} (D + \eta n K \sigma)^2. \quad (58)$$

This holds uniformly over the class and every permutation. Taking $\eta \downarrow 0$ proves the cap $\bar{L} D^2/2$. The equal-smoothness case has $\bar{L} = L$.

7 Optimization of the Step Size

We prove the two-sided scalar optimization used in the minimax corollary.

Lemma 2 (Two-branch optimization). *Let $n \geq 6$, $K \geq 1$, $T = nK$, and $L, D, \sigma > 0$. Define*

$$\Phi(\eta) = \frac{D^2}{\eta T} + \sigma^2 \min\{\eta, \eta^2 n L\}, \quad 0 < \eta \leq \frac{1}{6L}. \quad (59)$$

Then

$$\inf_{\eta} \Phi(\eta) \asymp \frac{LD^2}{T} + \min \left\{ \frac{\sigma D}{\sqrt{T}}, \left(\frac{L\sigma^2 D^4}{nK^2} \right)^{1/3} \right\}. \quad (60)$$

Proof. Put

$$A = \frac{LD^2}{T}, \quad S = \frac{\sigma D}{\sqrt{T}}, \quad R = \left(\frac{L\sigma^2 D^4}{nK^2} \right)^{1/3}.$$

For every admissible η , the deterministic term is at least $6A$. If $\eta \geq 1/(nL)$, the arithmetic-geometric mean inequality gives

$$\frac{D^2}{\eta T} + \eta \sigma^2 \geq 2S. \quad (61)$$

If $\eta \leq 1/(nL)$, minimizing $a/\eta + b\eta^2$ on the positive half-line, or weighted arithmetic-geometric mean, gives

$$\frac{D^2}{\eta T} + \eta^2 n L \sigma^2 \geq cR. \quad (62)$$

Hence every η gives $\Phi(\eta) \geq c'[A + \min\{S, R\}]$.

For the reverse inequality, define

$$\eta_S = \frac{D}{\sigma\sqrt{T}}, \quad \eta_R = \left(\frac{D^2}{n^2 K L \sigma^2} \right)^{1/3}. \quad (63)$$

The conditions $\eta_S \geq 1/(nL)$ and $\eta_R \leq 1/(nL)$ are respectively equivalent to

$$K \leq \frac{L^2 D^2 n}{\sigma^2} \quad \text{and} \quad K \geq \frac{L^2 D^2 n}{\sigma^2}. \quad (64)$$

Thus the candidate corresponding to $\min\{S, R\}$ lies in its correct branch. If that candidate is at most $1/(6L)$, substitution gives a constant multiple of S or R . The same substitution shows that its value dominates A , so it also upper bounds $A + \min\{S, R\}$ up to a constant.

If the candidate exceeds $1/(6L)$, choose $\eta = 1/(6L)$. This can only occur on the S branch. The condition $\eta_S > 1/(6L)$ implies

$$S < 6A, \quad \frac{\sigma^2}{6L} < 6A. \quad (65)$$

Therefore $\Phi(1/(6L)) < 12A$, which is again a constant multiple of $A + \min\{S, R\}$. This proves (60). $\square$

The heterogeneous optimization used in the main paper follows by the same argument. We record it explicitly.

Corollary 3 (Two-smoothness optimization). *Let $0 < \bar{L} \leq \hat{L}$ and $n\bar{L}/\hat{L} \geq 34$. Then*

$$\begin{aligned} & \inf_{0 < \eta \leq 1/(6\hat{L})} \left[\frac{D^2}{\eta n K} + \sigma^2 \min\{\eta, \eta^2 n \bar{L}\} \right] \\ & \asymp \frac{\hat{L} D^2}{n K} + \min \left\{ \frac{\sigma D}{\sqrt{n K}}, \left(\frac{\bar{L} \sigma^2 D^4}{n K^2} \right)^{1/3} \right\}. \end{aligned} \quad (66)$$

Proof. Repeat the proof of Lemma 2 with the noise curvature equal to $\bar{L}$ and the step endpoint set by $\hat{L}$. The deterministic term is at least $6\hat{L}D^2/(nK)$. The branch transition $1/(n\bar{L})$ is below $1/(6\hat{L})$. Hence the cubic candidate is always admissible. If the stochastic-gradient candidate exceeds the endpoint, use $\eta = 1/(6\hat{L})$. Exactly as before, both its deterministic and noise terms are then bounded by a constant multiple of $\hat{L}D^2/(nK)$. This proves (66). $\square$

Finally, for any constant a and function G ,

$$\inf_{\eta} \min\{a, G(\eta)\} = \min\{a, \inf_{\eta} G(\eta)\}. \quad (67)$$

Applying this identity to the fixed-step lower bound and Lemma 2 proves the minimax lower bound. The upper bound is Proposition 1 in the main paper.

8 Mechanical Constant Checks

As an additional check, direct enumeration of the exact finite sums in (11) and (12), as well as both common-coefficient inequalities, for every even $32 \leq n \leq 4096$. It reports

quantity	worst value
$A_i/(\ell n)$	0.3543616759
$B_i/n^{3/2}$	0.0003117429
negative-start coefficient gap	−0.0008678

The analytic bounds above cover all larger n . A separate random check samples 300,000 dimensionless fixed-step regimes and 20,000 optimization regimes to check the algebraic case reduction. These checks are not used as evidence for the theorem.

References

- Cha, J., Lee, J., and Yun, C. (2023). Tighter lower bounds for shuffling SGD: Random permutations and beyond. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 3855–3912. PMLR.
- Liu, Z. (2026). Random reshuffling dominates stochastic gradient descent. In *Proceedings of the 39th Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 4859–4882. PMLR.